

Show understanding of the basic Von Neumann model for a computer system and the stored program concept

9618_s25_qp_12_sc_1.a

QUESTION

- 1 (a) The table has six statements about the Von Neumann model for a computer system. Three of the statements are incorrect.

Statement number	Statement
1	The Program Counter (PC) stores the next instruction to be fetched from memory.
2	The Arithmetic and Logic Unit (ALU) performs mathematical and logical operations.
3	The Control Unit (CU) sends signals to other components on the data bus.
4	The Memory Data Register (MDR) transfers data to the memory address stored in the Memory Address Register (MAR).
5	The MAR stores an address from memory.
6	The Accumulator (ACC) stores the result of calculations.

Complete the table by writing the **three** incorrect statement numbers and the corrected statements.

Incorrect statement number	Corrected statement
.....	
.....	
.....	

[3]

MARK SCHEME

1(a)

1 mark for each correct row

3

Incorrect statement number	Corrected statement
1	The Program Counter (PC) stores the address of the next instruction to be fetched from memory.
3	The Control Unit (CU) sends signals to other components on the control bus.
4	The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR

9618_w23_qp_13_sc_9.a.i

QUESTION

- 9 A computer system is designed using the basic Von Neumann model.

Registers and buses are components in the Von Neumann model.

- (a) (i) Identify **three other** components in the Von Neumann model of a computer system.

Do not include registers or buses in your answers.

1

2

3

[3]

MARK SCHEME

9(a)(i)	1 mark for each bullet point (max 3).	3
	- Control unit (CU) - Arithmetic and Logic Unit (ALU) - Immediate Access Store (IAS) (System clock)	

QUESTION

- 5 (a) State what is meant by the **stored program concept** in the Von Neumann model of a computer system.

.....
..... [1]

MARK SCHEME

5(a)	Instructions and data are stored in the same memory space / in main memory.	1
------	--	----------

Show understanding of the purpose and role of registers, including the difference between general purpose and special purpose registers

9618_w25_qp_13_sc_6.b

QUESTION

(b) The Von Neumann model for a computer system uses registers.

Describe the role of the Memory Address Register (MAR) and Memory Data Register (MDR) in the fetch-execute (F-E) cycle.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

6(b)	1 mark for each bullet point, max 4 marks	4
	- MAR stores the <u>address</u> of the next instruction / data to be read from / or written to memory - The address is received from the Program Counter (PC) - The MDR stores the data / instruction in the address stored in the MAR which has been read / written - The instruction passes to the CIR for decoding and executing	

QUESTION

- (ii) Registers such as the Accumulator (ACC) and the Index Register (IX) are used in the CPU.

Identify **two** special purpose registers used in the CPU. Do **not** include the ACC or IX in your answers.

1

2

[2]

MARK SCHEME

3(a)(ii)	1 mark per bullet point, max 2 marks • Program counter (PC) • Memory Data Register (MDR) • Memory Address Register (MAR) • Current Instruction Register (CIR) • Status Register	2
----------	---	----------

QUESTION

- 6 (a) The processor uses several registers, including the Accumulator (ACC) and the Current Instruction Register (CIR).

Complete the table by describing the role of each register.

Register	Role
ACC	
CIR	

[2]

MARK SCHEME

6(a)	1 mark for each correct description, max 2 marks		2
	Register	Role	
	ACC	stores the intermediate results of arithmetic and logical operations // holds the result of a calculation	
	CIR	holds the instruction currently being decoded and/or executed	

QUESTION

- 1 (a) The table has six statements about the Von Neumann model for a computer system. Three of the statements are incorrect.

Statement number	Statement
1	The Program Counter (PC) stores the next instruction to be fetched from memory.
2	The Arithmetic and Logic Unit (ALU) performs mathematical and logical operations.
3	The Control Unit (CU) sends signals to other components on the data bus.
4	The Memory Data Register (MDR) transfers data to the memory address stored in the Memory Address Register (MAR).
5	The MAR stores an address from memory.
6	The Accumulator (ACC) stores the result of calculations.

Complete the table by writing the **three** incorrect statement numbers and the corrected statements.

Incorrect statement number	Corrected statement
.....	
.....	
.....	

[3]

MARK SCHEME

1(a)	1 mark for each correct row	3								
	<table><tr><th>Incorrect statement number</th><th>Corrected statement</th></tr><tr><td>1</td><td>The Program Counter (PC) stores the address of the next instruction to be fetched from memory.</td></tr><tr><td>3</td><td>The Control Unit (CU) sends signals to other components on the control bus.</td></tr><tr><td>4</td><td>The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR</td></tr></table>	Incorrect statement number	Corrected statement	1	The Program Counter (PC) stores the address of the next instruction to be fetched from memory.	3	The Control Unit (CU) sends signals to other components on the control bus.	4	The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR	
Incorrect statement number	Corrected statement									
1	The Program Counter (PC) stores the address of the next instruction to be fetched from memory.									
3	The Control Unit (CU) sends signals to other components on the control bus.									
4	The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR									

9618_s25_qp_12_sc_1.b

QUESTION

- (b) Registers that are used in the Fetch-Execute (F-E) cycle include the PC, MAR, MDR and the ACC.

Identify **one** other register and describe its role in the Fetch-Execute (F-E) cycle.

Register

Role

..... [2]

MARK SCHEME

1(b)	1 mark for name, 1 mark for corresponding role	2
	e.g. - Current Instruction Register // CIR - To store the instruction to be decoded / executed next - Status register // SR - To contain bits that can be referenced individually to indicate a state or event - Interrupt register - To store details of any interrupts that have occurred	

QUESTION

3 A student has a computer.

(a) The computer is designed using the Von Neumann model for a computer system.

Complete the table by describing the purpose of each of the given registers.

Register	Purpose
Program Counter (PC)	
Memory Address Register (MAR)	
Memory Data Register (MDR)	
Index Register (IX)	

[4]

MARK SCHEME

3(a)	1 mark for each correct purpose	4										
<table><tr><th>Register</th><th>Purpose</th></tr><tr><td>Program Counter (PC)</td><td>Stores the address of the next instruction to be fetched/executed</td></tr><tr><td>Memory Address Register (MAR)</td><td>Stores the address of the memory location where data will be read from/written to</td></tr><tr><td>Memory Data Register (MDR)</td><td>Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR</td></tr><tr><td>Index Register (IX)</td><td>Stores a number that will be added to the operand, to form the address of the data</td></tr></table>			Register	Purpose	Program Counter (PC)	Stores the address of the next instruction to be fetched/executed	Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to	Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR	Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data
Register	Purpose											
Program Counter (PC)	Stores the address of the next instruction to be fetched/executed											
Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to											
Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR											
Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data											

QUESTION

- 8 A computer designed using the Von Neumann model for a computer system contains general purpose registers and special purpose registers.

(a) Describe the purpose of the Status Register (SR).

.....

.....

.....

..... [2]

MARK SCHEME

8(a)	1 mark for each bullet point (max 2) • To store the value of flags/bits • ... that can be changed/set/cleared after arithmetic / logical operations • To allow flags to be checked • ... to change instruction sequence	2
------	--	----------

QUESTION

(b) Identify **two** differences between general purpose registers and special purpose registers.

1

.....

.....

2

.....

..... [2]

MARK SCHEME

8(b)	1 mark for each bullet point (max 2) - Special purpose registers have a specified role in the machine whereas general purpose registers can be used for all purposes defined by the programmer - Special purpose registers hold the state of the program's execution while general purpose registers hold the program's data during operations	2
------	---	----------

9618_w23_qp_13_sc_9.a.ii

QUESTION

(ii) Identify **two** differences between special purpose registers and general purpose registers.

1

.....

2

.....

[2]

MARK SCHEME

9(a)(ii)	1 mark for each bullet point (max 2). - Special purpose registers hold the status of a program whereas general purpose registers hold the temporary data while performing operations. - Special purpose registers are specialised for a specific use, whereas general purpose registers are used for any purpose. - General purpose registers can be used by most instructions, whereas special purpose can only be used by certain instruction	2
----------	---	----------

QUESTION

- 5 The Central Processing Unit (CPU) of the basic Von Neumann model for a computer system contains several special purpose registers.

- (a) The Memory Data Register (MDR), Index Register (IX) and the Accumulator (ACC) are examples of special purpose registers.

Identify **two other** special purpose registers **and** state their role in the CPU.

Special purpose register 1

Role

.....

.....

Special purpose register 2

Role

.....

.....

[4]

MARK SCHEME

5(a)	1 mark for identification of the register and 1 mark for role (max 2 for each register) • Program Counter (PC) • stores the <u>address</u> where the <u>next</u> instruction is to be read from • Memory Address Register (MAR) • stores the address of the memory location (or an I/O component) currently being read from or written to • Current Instruction Register (CIR) • holds the instruction currently being decoded and/or executed • Status Register • Contains bits which can be referenced individually and set or cleared depending on the operation e.g. overflow, underflow	4
------	--	----------

QUESTION

- 4 (a) A Central Processing Unit (CPU) contains several special purpose registers and other components.

- (i) State the roles of the following registers.

Memory Address Register (MAR)

.....

.....

Memory Data Register (MDR)

.....

.....

[2]

MARK SCHEME

4(a)(i)	1 mark for each register: MAR: holds address in memory from which data will be read / to which data will be written MDR: holds the data/instructions which has been read from or is to be written to the address in the MAR	2
---------	---	----------

QUESTION

(b) A Central Processing Unit (CPU) contains several special purpose registers and other components.

(i) State the role of the following registers.

Program Counter (PC)

.....

.....

Index Register (IX)

.....

.....

Status Register (SR)

.....

.....

[3]

MARK SCHEME

5(b)(i)	1 mark for each special purpose register: Program Counter (PC): - to store the address / location / memory location of the <u>next</u> instruction to be fetched Index Register (IX): - to store a value that is added to an address to give another address Status Register (SR): - to store flags which are set by events // from the results of arithmetic and logic operations and interrupt flags	3
---------	--	----------

QUESTION

8 The Von Neumann model for a computer system uses registers.

(a) Describe the role of the following special purpose registers in the fetch-execute (F-E) cycle.

(i) Memory Address Register (MAR)

.....

.....

.....

Memory Data Register (MDR)

.....

.....

.....

[4]

MARK SCHEME

8(a)(i)	1 mark for each bullet point to max 2 for each register MAR • Stores the next <u>address</u> to be fetched • ... held in the Program Counter (PC) • The data at this address is then fetched MDR • Stores the data from the address pointed to by the MAR • The data in it is copied to the Current Instruction Register (CIR)	4
---------	--	----------

QUESTION

(ii) Another special purpose register is the Index Register.

Identify **one other** special purpose register used in the Von Neumann model for a computer system.

.....

..... [1]

MARK SCHEME

8(a)(ii)	1 mark for a correct register e.g. Program Counter (PC) Current Instruction Register (CIR) Status register Interrupt register	1
----------	---	----------

9618_s21_qp_11_sc_3.a

QUESTION

- 3** A processor has one general purpose register, the Accumulator (ACC), and several special purpose registers.

- (a)** Complete the following description of the role of the registers in the fetch-execute cycle by writing the missing registers.

The holds the address of the next instruction to be loaded. This address is sent to the

The holds the data fetched from this address.

This data is sent to the and the Control Unit decodes the instruction's opcode.

The is incremented.

[5]

MARK SCHEME

3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register . The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5
------	--	----------

Show understanding of the purpose and roles of the Arithmetic and Logic Unit (ALU), Control Unit (CU) and system clock, Immediate Access Store (IAS)

9618_s25_qp_13_sc_2.a.i

QUESTION

2 A computer has a processor.

(a) The processor has a Control Unit (CU) and system clock.

(i) Explain how the CU and the system clock work together.

.....

.....

.....

..... [2]

MARK SCHEME

2(a)(i)	1 mark each to max 2 - max 1 for each set of marks - The control unit synchronises the actions of the processor - by sending a command / signal on each timing signal produced by the system clock - using / along the control bus	2
---------	--	----------

QUESTION

3 A computer system has a dual-core Central Processing Unit (CPU).

(a) State the purpose of the system clock and the Control Unit (CU) in a CPU.

System clock

.....

CU

.....

[2]

MARK SCHEME

3(a)	1 mark for system clock and 1 mark for Control Unit System clock • To synchronise operations • ... by creating and transmitting timing signals on the control bus Control Unit • Sends/receives control signals along control bus • Reads an instruction from the contents of the memory location whose address is stored in PC • Coordinates/synchronises the activity of other components in the CPU • Manages the execution of instructions • Controls communication between the components in the CPU	2
------	---	----------

QUESTION

(b) Describe what is meant by the **Immediate Access Store (IAS)** in a computer system.

.....

.....

.....

..... [2]

MARK SCHEME

5(b)	1 mark for each bullet point (max 2) - Immediate Access Store holds all the data / instructions / programs currently in use - Immediate Access Store is volatile memory - Immediate Access Store has fast access times	2
------	--	----------

9618_s23_qp_13_sc_7.b

QUESTION

(b) The computer has a Control Unit (CU), system clock and control bus.

Explain how the CU, system clock and control bus operate to transfer data between the components of the computer system.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

7(b)	1 mark each to max 4 - The system clock gives out timing signals ... which are sent on the control bus ...to synchronise the other system components - The Control Unit initiates data transfer ...by generating signals that are sent on the control bus to other components	4
------	---	----------

9618_w22_qp_13_sc_4.b

QUESTION

(b) A computer system contains a system clock.

Describe the purpose of the system clock.

.....

.....

.....

..... [2]

MARK SCHEME

4(b)	1 mark for each bullet point (max 2): • synchronise operations • ... by creating timing signals • to keep track of the date and time / timestamp files • to process operations in the correct order / sequence	2
------	--	----------

9618_w22_qp_11_sc_5.b.iii

QUESTION

(iii) Describe the purpose of the Control Unit (CU) in a CPU.

.....

.....

.....

..... [2]

MARK SCHEME

5(b)(iii)	1 mark for each bullet point (max 2): • to coordinate / synchronise the actions of other components in the CPU • to send / receive control signals along the control bus • to manage the execution of instructions (in sequence) • to control the communication between the components of the CPU	2
-----------	---	----------

QUESTION

5 Seth uses a computer for work.

(a) Complete the following descriptions of internal components of a computer by writing the missing terms.

The transmits the signals to coordinate events based on the electronic pulses of the

The carries data to the components, while the carries the address where data needs to be written to or read from.

The performs mathematical operations and logical comparisons.

[5]

MARK SCHEME

5(a)	1 mark for each term correctly inserted The control unit/bus transmits the signals to coordinate events based on the pulses of the (system) clock. The data bus carries data to components, while the address bus carries the address where data is being written to or read from. The arithmetic logic unit/ALU performs mathematical operations and logical comparisons.	5
------	--	----------

Show understanding of how data are transferred between various components of the computer system using the address bus, data bus and control bus

9618_s25_qp_11_sc_6.a.iii

QUESTION

(iii) The two computers have different bus widths.

Explain how different bus widths affect the performance of a computer.

.....

.....

.....

..... [2]

MARK SCHEME

6(a)(iii)	1 mark each to max 2 e.g. • Wider data bus means more data can be transferred between components at a time • there is less delay / latency when fetching data for a running process • Wider address bus means larger memory addresses can be used • allowing more memory locations to be accessed directly • so less likely to run out of memory	2
-----------	--	----------

QUESTION

- (ii) Describe the roles of the address bus, the data bus **and** buffers in the process of writing data to the optical disc reader/writer.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

8(c)(ii)	1 mark for each component (max 3) Buffers - A buffer temporarily holds data until it is ready to be transmitted to the device Address Bus - The address of the data to be written to the device (in RAM) is carried on the address bus Data Bus - All data to be written to the device / buffer is carried on the data bus	3
----------	--	----------

QUESTION

(b) The computer has a Control Unit (CU), system clock and control bus.

Explain how the CU, system clock and control bus operate to transfer data between the components of the computer system.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

7(b)	1 mark each to max 4 • The system clock gives out timing signals • ... which are sent on the control bus • ...to synchronise the other system components • The Control Unit initiates data transfer • ...by generating signals that are sent on the control bus to other components	4
------	--	----------

QUESTION

5 A student has purchased a new laptop.

(a) The laptop is designed using the Von Neumann model for a computer system.

Identify **two** types of signal that a control bus can transfer.

1

2 [2]

MARK SCHEME

5(a)	1 mark each to max 2 Examples: • Interrupt • Timing • Read • Write	2
------	---	----------

9618_w22_qp_11_sc_5.b.ii

QUESTION

(ii) Tick (✓) **one** box in each row to identify the system bus used by each CPU component.

CPU component	Data bus	Address bus	Control bus
System clock			
Memory Address Register (MAR)			

[1]

MARK SCHEME

5(b)(ii)	1 mark for both rows:			1
	CPU component	Data bus	Address bus	Control bus
	System clock			✓
	Memory Address Register (MAR)		✓	

QUESTION

5 Seth uses a computer for work.

(a) Complete the following descriptions of internal components of a computer by writing the missing terms.

The transmits the signals to coordinate events based on the electronic pulses of the

The carries data to the components, while the carries the address where data needs to be written to or read from.

The performs mathematical operations and logical comparisons.

[5]

MARK SCHEME

5(a)	1 mark for each term correctly inserted The control unit/bus transmits the signals to coordinate events based on the pulses of the (system) clock. The data bus carries data to components, while the address bus carries the address where data is being written to or read from. The arithmetic logic unit/ALU performs mathematical operations and logical comparisons.	5
------	--	----------

Show understanding of how factors contribute to the performance of the computer system

9618_w25_qp_12_sc_3.c

QUESTION

(c) A student buys a new computer. The table shows the specifications of the old computer and the new computer.

Old computer	New computer
1.8GHz dual core processor	2.3GHz dual core processor
16MB cache	32MB cache

Explain why increasing the clock speed **and** increasing the cache memory will improve the performance of the computer.

Clock speed

 Cache memory

 [4]

MARK SCHEME

3(c)	1 mark per bullet point, max 2 marks for each Clock Speed: - Processor can perform more F-E cycles per second ... so more instructions / data can be processed each second Cache Memory: - Can store more of the most frequently used instructions ... which reduces the need to access slower RAM	4
------	---	----------

QUESTION

(b) Increasing the number of cores in a processor can affect the performance of a computer.

Describe the drawbacks of increasing the number of cores in a processor.

.....

.....

.....

..... [2]

MARK SCHEME

6(b)	1 mark per bullet point, max 2 marks e.g. • Latency may be increased • ... because the cores must communicate with one another • There is a potential for dead-lock situations • ... where one core may wait for information from other cores which in turn are waiting for the first one • Not all software is designed to use multi-cores • ... so some of the additional cores would be idle • Increased heat generation • ... which could cause damage to other components	2
------	---	----------

QUESTION

(b) The computer has cache memory.

Describe **one** benefit of the computer using cache memory.

.....

.....

.....

..... [2]

MARK SCHEME

2(b)	1 mark each • Using cache memory improves system performance • because cache is fast access memory close to the CPU • which stores frequently used instructions / data • so that they can be accessed faster than from RAM	2
------	--	----------

9618_s25_qp_11_sc_6.a.ii

QUESTION

(ii) The two computers have different amounts of RAM.

Explain how different amounts of RAM affect the performance of a computer.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(a)(ii)	1 mark each to max 3 • More RAM means more currently running data and instructions can be stored • without needing to use virtual memory • without having to fetch the data from secondary storage first • which has a slower access time • Less latency / delay waiting for instructions / data	3
----------	---	----------

QUESTION

(iii) The two computers have different bus widths.

Explain how different bus widths affect the performance of a computer.

.....

.....

.....

..... [2]

MARK SCHEME

6(a)(iii)	1 mark each to max 2 e.g. • Wider data bus means more data can be transferred between components at a time • there is less delay / latency when fetching data for a running process • Wider address bus means larger memory addresses can be used • allowing more memory locations to be accessed directly • so less likely to run out of memory	2
-----------	--	----------

QUESTION

(b) (i) The number of cores in the processor affects the performance of the computer system.

Identify **one other** feature of a processor that can affect the performance of a computer system **and** state why it affects the performance.

Feature

.....

Reason

..... [2]

MARK SCHEME

3(b)(i)	1 mark for the feature, 1 mark max for the matching reason e.g. Feature: clock speed - Higher clock speed means that more F-E cycles are executed per second // Higher clock speed results in more throughput Feature: bus width - Larger bus width means that more data transferred at the same time	2
---------	--	----------

9618_w23_qp_11_sc_5.c.i

QUESTION

(c) A computer has a single 2.1 GHz CPU.

(i) Describe how increasing the clock speed to 4 GHz can increase the performance of the computer.

.....

.....

..... [1]

MARK SCHEME

5(c)(i)	1 mark for each bullet point (max 1) - The CPU can now perform nearly twice as many F-E cycles per second - Instead of 2.1 billion F-E cycles per second, the CPU can now perform 4 billion FE cycles per second	1
---------	---	----------

QUESTION

- (ii) A second computer has a CPU with two 2.1 GHz cores.

Explain why the second computer does not always run twice as fast as the computer with one 2.1 GHz CPU.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

5(c)(ii)	1 mark for each bullet point (max 5) • Multiple cores introduce additional overheads • ...because of the need for communication between cores • Software may not be designed for multiple cores... • ...so one of the cores will be left idle • Memory access speed may not match speed of cores... • ...so causing delay • The two computers may have more differences than just the cores • ...one may have more RAM which allows faster multitasking • ...one may have a GPU • ...etc.	5
----------	---	----------

QUESTION

- (b) Describe **two** ways the hardware of a laptop can be upgraded to improve the performance **and** explain how each upgrade improves the performance.

1

.....

.....

.....

.....

.....

.....

2

.....

.....

.....

.....

.....

.....

[4]

MARK SCHEME

5(b)	1 mark for description; 1 mark for corresponding explanation Examples • Increase number of cores • Each core can independently carry out a process at the same time // so that more instructions are performed in parallel • Increase RAM capacity • ... allowing more applications to reside in memory at the same time, saving disk access times • Increase cache memory • More data can be stored in fast access so less time is spent accessing from RAM • Increase clock speed • More Fetch-Decode-Execute (FDE) cycles can run each second / per unit time	4
------	--	----------

QUESTION

- (c) Upgrading secondary storage to solid state typically improves the performance of computer systems.

Identify **one other** upgrade to the hardware **and** explain why it improves the performance of a computer system.

Upgrade

Explanation

.....

.....

[2]

MARK SCHEME

4(c)	1 mark for identification of a correct upgrade: 1 mark for a corresponding explanation: Examples: • increase quantity of RAM • ... so allowing more applications to reside in memory at the same time, saving disk access times • increase the size of cache memory • ... so that the CPU can continue working without waiting for data • increase clock speed • ... so that more instructions are performed in a time period • increase the number of processors / cores • ... so that more instructions are performed in parallel	2
------	---	----------

QUESTION

- (d) Explain the reasons why increasing the amount of cache memory can improve the performance of a CPU.

.....

.....

.....

..... [2]

MARK SCHEME

7(d)	1 mark for each bullet point (max 2): • cache is fast access memory (close to the CPU) • cache stores frequently used instructions / data • ... more cache means more instructions / data can be transferred faster • ... less swapping between RAM and cache • prevents the CPU idling while waiting for data	2
------	--	----------

9618_s21_qp_12_sc_5.b

QUESTION

(b) Describe the ways in which the following factors can affect the performance of his laptop computer.

Number of cores

.....

.....

.....

.....

Clock speed

.....

.....

.....

.....

[4]

MARK SCHEME

5(b)	1 mark per bullet point to max 3 per factor. max 4 overall. Number of cores: • Each core processes one <u>instruction</u> per clock pulse • More/multiple cores mean that sequences of instructions can be split between them • ... and so more than one <u>instruction</u> is executed per clock pulse // more sequences of instructions can be run at the same time • More cores decreases the time taken to complete task Clock speed: • Each <u>instruction</u> is executed on a clock pulse // one F-E cycle is run on each clock pulse • ... so the clock speed dictates the number of <u>instructions</u> that can be run per second • The faster the clock speed the more <u>instructions</u> can be run per second	4
------	---	----------

Understand how different ports provide connection to peripheral devices

9618_s25_qp_13_sc_2.c

QUESTION

- (c) The computer connects to a monitor using a High Definition Multimedia Interface (HDMI) cable that connects into an HDMI port.

Explain how HDMI provides connection to peripheral devices.

.....

.....

.....

..... [2]

MARK SCHEME

2(c)	1 mark each to max 2 e.g. • HDMI transfers both audio and video using a single cable • HDMI has a high bandwidth • Data is transmitted in a stream of uncompressed digital signals • HDMI uses a technology called Transition-Minimized Differential Signalling (TMDS)	2
------	--	----------

QUESTION

- (b) The student needs to connect the computer to a monitor that has a screen resolution of 2560×1600 pixels. The monitor also has built-in speakers.

The computer has a Video Graphics Array (VGA) port and a High Definition Multimedia Interface (HDMI) port.

Explain the benefits of connecting the monitor to the computer using the HDMI port instead of the VGA port.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

3(b)	1 mark for each bullet point (max 4) e.g. • HDMI has faster transfer rates than VGA • ... needed due to high resolution / large number of pixels of monitor // HDMI supports the high resolution of the monitor • HDMI supports video and audio transfer between computer and monitor speakers • ... so no separate sound cable is needed unlike VGA • HDMI is digital interface therefore no data is lost in transfer to analogue and back • HDMI is less prone to error/crosstalk/external interference	4
------	--	----------

QUESTION

- (ii) A solid state (flash) memory drive is automatically recognised by the computer when it is plugged into a port in the computer.

Identify an appropriate type of port to connect the solid state memory drive to the computer.

Explain how this port provides an automatic connection.

Port

Explanation

.....

.....

.....

[3]

MARK SCHEME

3(b)(ii)	1 mark for correct port. 2 marks for explanation Port: USB / Universal Serial Bus Explanation: • A voltage change occurs when the drive is plugged in • The computer detects this voltage change • The code of the device is transferred to computer • ... the OS finds the code of the device in the list of devices • ... and loads the appropriate device driver	3
----------	--	----------

QUESTION

- (d) Identify **one** port that could be used to connect the virtual reality headset to the laptop.

Justify your choice.

Port

Justification

.....

.....

.....

[3]

MARK SCHEME

7(d)	1 mark for naming a correct port 2 marks for matching justification • USB • ...has fast data transfer speeds for data (to the headset) • ...is a universal/popular cable // universal standard • HDMI • ...allows video and audio to be transferred (on the same cable) • ...convenience of HDMI as no need for two cables	3
------	---	----------

9618_w23_qp_12_sc_8.c.i

QUESTION

(c) An optical disc reader/writer is connected to the computer.

(i) Give the name of **one** port that can provide a connection for the optical disc reader/writer.

..... [1]

MARK SCHEME

8(c)(i)	1 mark for each bullet point (max 1) • USB / Universal Serial Bus • HDMI	1
---------	---	----------

9618_s23_qp_12_sc_5.c.i

QUESTION

(c) Peripherals are connected to the laptop using ports.

(i) A printer is connected to a Universal Serial Bus (USB) port.

Describe how data is transmitted through a USB port.

.....

.....

.....

..... [1]

MARK SCHEME

5(c)(i)	1 mark for a correct answer 1 bit is transferred at a time - Can be synchronous or asynchronous - USB-3 is full duplex and earlier versions are half-duplex	1
---------	--	----------

9618_s23_qp_12_sc_5.c.ii

QUESTION

(ii) A monitor is connected to the laptop using a different type of port.

Identify **one** other type of port that can be used to connect the monitor.

..... [1]

MARK SCHEME

5(c)(ii)	1 mark for identification of a suitable port Examples - HDMI - DisplayPort	1
----------	--	----------

9618_w22_qp_11_sc_7.e

QUESTION

(e) State the name of a peripheral device port that provides a physical connection in the computer for each of these peripherals.

3D printer

Monitor [2]

MARK SCHEME

7(e)	1 mark for each device. 3D printer: USB port / COM port Monitor: HDMI / VGA / USB / DisplayPort	2
------	--	----------

Describe the stages of the Fetch-Execute (F-E) cycle

9618_w25_qp_13_sc_6.b

QUESTION

(b) The Von Neumann model for a computer system uses registers.

Describe the role of the Memory Address Register (MAR) and Memory Data Register (MDR) in the fetch-execute (F-E) cycle.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

6(b)	1 mark for each bullet point, max 4 marks - MAR stores the <u>address</u> of the next instruction / data to be read from / or written to memory - The address is received from the Program Counter (PC) - The MDR stores the data / instruction in the address stored in the MAR which has been read / written - The instruction passes to the CIR for decoding and executing	4
------	--	----------

QUESTION

(ii) The processor runs the Fetch-Execute (F-E) cycle.

Write the stages of the F-E cycle using register transfer notation.

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

2(a)(ii)	1 mark each • $MAR \leftarrow [PC]$ and $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$ • Correct order • For example: • $MAR \leftarrow [PC]$ • $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$	4
----------	--	----------

QUESTION

(b) Registers that are used in the Fetch-Execute (F-E) cycle include the PC, MAR, MDR and the ACC.

Identify **one** other register and describe its role in the Fetch-Execute (F-E) cycle.

Register

Role

..... [2]

MARK SCHEME

1(b)	1 mark for name, 1 mark for corresponding role e.g. • Current Instruction Register // CIR • To store the instruction to be decoded / executed next • Status register // SR • To contain bits that can be referenced individually to indicate a state or event • Interrupt register • To store details of any interrupts that have occurred	2
------	---	----------

9618_w23_qp_13_sc_9.b

QUESTION

(b) The following incomplete table contains steps of the Fetch-Execute (F-E) cycle and their descriptions.

Complete the table by writing the missing steps using register transfer notation **and** the missing descriptions.

Step	Description
.....	The address in PC is incremented.
$MDR \leftarrow [[MAR]]$	
$MAR \leftarrow [PC]$	
.....	The contents of MDR are copied into CIR.

[4]

MARK SCHEME

9(b)

1 mark for each correct answer (shaded cells)

4

Step	Description
$PC \leftarrow [PC] + 1$	The address in PC is incremented.
$MDR \leftarrow [[MAR]]$	The data in the location pointed to by the MAR is copied to the MDR.
$MAR \leftarrow [PC]$	The contents of PC are copied to the MAR.
$CIR \leftarrow [MDR]$	The contents of MDR are copied into CIR.

9618_s23_qp_13_sc_7.c

QUESTION

- (c) Complete the table by writing the register transfer notation for each stage of the Fetch-Execute (F-E) cycle given in the table.

Stage description	Register transfer notation
The Program Counter (PC) is incremented	
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	

[2]

MARK SCHEME

7(c)	1 mark for each Register transfer notation	2						
<table><tr><th>Stage description</th><th>Register transfer notation</th></tr><tr><td>The Program Counter (PC) is incremented</td><td>$PC \leftarrow [PC] + 1$</td></tr><tr><td>The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)</td><td>$MDR \leftarrow [[MAR]]$</td></tr></table>			Stage description	Register transfer notation	The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$	The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$
Stage description	Register transfer notation							
The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$							
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$							

QUESTION

- (ii) State when interrupts are detected during the Fetch-Execute (F-E) cycle.

.....
 [1]

MARK SCHEME

4(a)(ii)	after completion of the execute stage // before the cycle begins	1
----------	--	---

QUESTION

- (c) Write the register transfer notation for each of the stages in the fetch-execute cycle described in the table.

Description	Register transfer notation
Copy the address of the next instruction into the Memory Address Register.	
Increment the Program Counter.	
Copy the contents of the Memory Data Register into the Current Instruction Register.	

[3]

MARK SCHEME

7(c)	1 mark for each correct row:	3								
<table><tr><th>Description</th><th>Register transfer notation</th></tr><tr><td>Copy the address of the next instruction into the Memory Address Register.</td><td>MAR ← [PC]</td></tr><tr><td>Increment the Program Counter.</td><td>PC ← [PC] + 1</td></tr><tr><td>Copy the contents of the Memory Data Register into the Current Instruction Register.</td><td>CIR ← [MDR]</td></tr></table>		Description	Register transfer notation	Copy the address of the next instruction into the Memory Address Register.	MAR ← [PC]	Increment the Program Counter.	PC ← [PC] + 1	Copy the contents of the Memory Data Register into the Current Instruction Register.	CIR ← [MDR]	
Description	Register transfer notation									
Copy the address of the next instruction into the Memory Address Register.	MAR ← [PC]									
Increment the Program Counter.	PC ← [PC] + 1									
Copy the contents of the Memory Data Register into the Current Instruction Register.	CIR ← [MDR]									

QUESTION

8 The Von Neumann model for a computer system uses registers.

(a) Describe the role of the following special purpose registers in the fetch-execute (F-E) cycle.

(i) Memory Address Register (MAR)

.....

.....

.....

Memory Data Register (MDR)

.....

.....

.....

[4]

MARK SCHEME

8(a)(i)	1 mark for each bullet point to max 2 for each register MAR • Stores the next <u>address</u> to be fetched • ... held in the Program Counter (PC) • The data at this address is then fetched MDR • Stores the data from the address pointed to by the MAR • The data in it is copied to the Current Instruction Register (CIR)	4
---------	--	----------

QUESTION

6 (a) There are **two** errors in the following register transfer notation for the fetch-execute cycle.

```

1  MAR ← [PC]
2  PC ← [PC] - 1
3  MDR ← [MAR]
4  CIR ← [MDR]

```

Complete the following table by:

- identifying the line number of each error
- describing the error
- writing the correct statement.

Line number	Description of the error	Correct statement

[4]

MARK SCHEME

6(a)	1 mark for identification of line and description of error 1 mark for the correct statement <table> <tr> <th>Line number</th><th>Description of the error</th><th>Correct statement</th></tr> <tr> <td>2</td><td>Program Counter should be incremented, not decremented</td><td>$PC \leftarrow [PC] + 1$</td></tr> <tr> <td>3</td><td>It should be the contents of the address in the MAR</td><td>$MDR \leftarrow [[MAR]]$</td></tr> </table>	Line number	Description of the error	Correct statement	2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$	3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$	4
Line number	Description of the error	Correct statement									
2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$									
3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$									

QUESTION

- 3 A processor has one general purpose register, the Accumulator (ACC), and several special purpose registers.

- (a) Complete the following description of the role of the registers in the fetch-execute cycle by writing the missing registers.

The holds the address of the next instruction to be loaded. This address is sent to the

The holds the data fetched from this address.

This data is sent to the and the Control Unit decodes the instruction's opcode.

The is incremented.

[5]

MARK SCHEME

3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register. The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5
------	--	----------

Show understanding of the purpose of interrupts

9618_s25_qp_12_sc_1.c

QUESTION

(c) Explain how an interrupt from an input device will be detected and handled in the F-E cycle.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

1(c)	1 mark each to max 4 • At the start / end of FE cycle the interrupt register is checked • The priority of any interrupts waiting is checked • if the priority of the interrupt is higher than the current process • the contents of the registers are stored on the stack • The relevant Interrupt Service Routine (ISR) / interrupt handler is called to process the interrupt • When the ISR has finished, a further check is made for higher priority interrupts • if no more interrupts of higher priority, the register contents are restored and the next FE cycle continues	4
------	---	----------

9618_w23_qp_13_sc_9.c

QUESTION

(c) Interrupts can be caused by software programs or hardware devices.

State **one** cause of a software interrupt.

.....

..... [1]

MARK SCHEME

9(c)	1 mark for an appropriate example. Examples: • Division by zero • Runtime error • Out of memory bounds • Program requesting an external device / input • Buffer overflow	1
------	--	----------

9618_w23_qp_13_sc_9.d

QUESTION

- (d) The following statements describe the stages that the CPU performs when an interrupt is detected.

There are **three** missing statements.

Write the letter of the missing statements from the table in the correct place to complete the description.

- 1 At the end of each Fetch-Execute (F-E) cycle, the processor checks if an interrupt flag is set.
- 2
- 3 If the interrupt priority is high enough, the processor saves the current contents of the registers.
- 4
- 5 When servicing of the interrupt is complete, the processor restores the registers.
- 6

Letter	Stage
A	The address of the Interrupt Service (ISR) handling routine is loaded into the Program Counter (PC).
B	Lower priority interrupts are re-enabled.
C	The device causing the interrupt transfers data to the CPU.
D	The processor identifies the source of the interrupt and checks the priority of the interrupt.
E	The ISR is incremented.

[3]

MARK SCHEME

9(d)	1 mark for each letter in the correct position (2, 4 and 6) D A B	3
------	--	----------

9618_s23_qp_11_sc_6

QUESTION

- 6** An interrupt is generated when a key is pressed on a computer keyboard.

Explain how the computer handles this interrupt.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

6	1 mark each to max 5 • An interrupt flag is raised in the (interrupt) register • At the end of the current FE cycle // at the start of the next FE cycle • The system checks the interrupt register for higher priority interrupts than current process • If true, it stores the current contents of the registers on the stack • The appropriate interrupt service routine (ISR) for the key press is called • The input data from the keyboard is processed • The contents of the registers are restored from the stack • ... and control is passed back to previous process	5
----------	--	----------

9618_w22_qp_13_sc_4.a.ii

QUESTION

(ii) State when interrupts are detected during the Fetch-Execute (F-E) cycle.

.....
..... [1]

MARK SCHEME

4(a)(ii)	after completion of the execute stage // before the cycle begins	1
----------	--	---

9618_w22_qp_11_sc_5.c

QUESTION

(c) Describe the purpose of an interrupt in a computer system.

.....
.....
.....
..... [2]

MARK SCHEME

5(c)	1 mark for each bullet point: - to send a signal from a device or process ... seeking the attention of the processor	2
------	--	----------

9618_w22_qp_11_sc_5.d

QUESTION

(d) Identify **two** causes of a software interrupt.

1
.....
2
..... [2]

MARK SCHEME

5(d)	1 mark for each bullet point (max 2). For example: • division by zero // runtime error in a program • attempt to access an invalid memory location • array index out of bounds • stack overflow	2
------	---	----------

9618_w21_qp_12_sc_7.a

QUESTION

7 A computer has system software.

(a) The Operating System handles interrupts.

Tick (✓) **one** box in each row to identify whether each event is an example of a hardware interrupt or a software interrupt.

Event	Hardware interrupt	Software interrupt
Buffer full		
Printer is out of paper		
User has pressed a key on the keyboard		
Division by zero		
Power failure		
Stack overflow		

[3]

MARK SCHEME

7(a)

1 mark per pair of rows (shaded & unshaded)

3

Event	Hardware Interrupt	Software Interrupt	
Buffer full		✓	} 1
Printer is out of paper	✓		
User has pressed a key on the keyboard	✓		} 1
Division by zero		✓	
Power failure	✓		} 1
Stack overflow		✓	